MPG No. 7.1.20 $mm2$ [Type I, orthorhombic] [G tensor]

* Rank 0 tensor. * Rank 1 tensor. * Rank 2 tensor (s).

$$\begin{bmatrix} 0 & G_{xy} & 0 \\ G_{xy} & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

$$G_{xy} = \mathbb{G}^{(1)}_{xy}$$

* Rank 2 tensor (a).

$$\begin{bmatrix} 0 & G_{xy} & 0 \\ -G_{xy} & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

$$G_{xy} = \mathbb{Q}^{(1)}_{z}$$

* Rank 3 tensor (ss).

$$\begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \\ G_{yzx} & 0 & 0 \\ 0 & G_{yzx} & 0 \\ 0 & 0 & G_{yzx} \end{bmatrix}$$

$$G_{yzx} = \mathbb{G}^{(1)}_{xyz}$$

* Rank 3 tensor (sa).

$$\begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \\ G_{yzx} & 0 & 0 \\ 0 & G_{zxy} & 0 \\ 0 & 0 & -G_{yzx} - G_{zxy} \end{bmatrix}$$

$$G_{yzx} = 3\mathbb{Q}^{(1)}_{u} + \mathbb{Q}^{(1)}_{v}$$
$$G_{zxy} = -3\mathbb{Q}^{(1)}_{u} + \mathbb{Q}^{(1)}_{v}$$

* Rank 3 tensor (a).

$$\begin{bmatrix} G_{yzx} & 0 & 0 \\ 0 & G_{zxy} & 0 \\ 0 & 0 & G_{xyz} \end{bmatrix}$$

$$G_{yzx} = \mathbb{Q}^{(1)}_{0} - \mathbb{Q}^{(2)}_{u} + \mathbb{Q}^{(2)}_{v}$$
$$G_{zxy} = \mathbb{Q}^{(1)}_{0} - \mathbb{Q}^{(2)}_{u} - \mathbb{Q}^{(2)}_{v}$$
$$G_{xyz} = \mathbb{Q}^{(1)}_{0} + 2\mathbb{Q}^{(2)}_{u}$$

* Rank 4 tensor (sss).

$$\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & G_{xxxy} \\ 0 & 0 & 0 & 0 & 0 & G_{yyxy} \\ 0 & 0 & 0 & 0 & 0 & G_{zzxy} \\ 0 & 0 & 0 & 0 & G_{yzzx} & 0 \\ 0 & 0 & 0 & G_{yzzx} & 0 & 0 \\ G_{xxxy} & G_{yyxy} & G_{zzxy} & 0 & 0 & 0 \end{bmatrix}$$

$$G_{xxxy} = \mathbb{G}_{4z}^{\alpha(1)} - \mathbb{G}_{4z}^{\beta(1)} + \mathbb{G}_{xy}^{(1)} + 2\mathbb{G}_{xy}^{(2)}$$
$$G_{yyxy} = -\mathbb{G}_{4z}^{\alpha(1)} - \mathbb{G}_{4z}^{\beta(1)} + \mathbb{G}_{xy}^{(1)} + 2\mathbb{G}_{xy}^{(2)}$$
$$G_{zzxy} = 2\mathbb{G}_{4z}^{\beta(1)} + \mathbb{G}_{xy}^{(1)}$$
$$G_{yzzx} = 2\mathbb{G}_{4z}^{\beta(1)} + \mathbb{G}_{xy}^{(2)}$$

* Rank 4 tensor (ssa).

$$\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & G_{xxxy} \\ 0 & 0 & 0 & 0 & 0 & G_{yyxy} \\ 0 & 0 & 0 & 0 & 0 & G_{zzxy} \\ 0 & 0 & 0 & 0 & G_{yzzx} & 0 \\ 0 & 0 & 0 & -G_{yzzx} & 0 & 0 \\ -G_{xxxy} & -G_{yyxy} & -G_{zzxy} & 0 & 0 & 0 \end{bmatrix}$$

$$G_{xxxy} = \mathbb{G}_{xy}^{(3)} + 2\mathbb{Q}_{z}^{(1)} - 2\mathbb{Q}_{z}^{\alpha(1)} + 2\mathbb{Q}_{z}^{\beta(1)}$$
$$G_{yyxy} = \mathbb{G}_{xy}^{(3)} - 2\mathbb{Q}_{z}^{(1)} + 2\mathbb{Q}_{z}^{\alpha(1)} + 2\mathbb{Q}_{z}^{\beta(1)}$$
$$G_{zzxy} = \mathbb{G}_{xy}^{(3)} - 4\mathbb{Q}_{z}^{\beta(1)}$$
$$G_{yzzx} = -\mathbb{Q}_{z}^{(1)} - 4\mathbb{Q}_{z}^{\alpha(1)}$$

* Rank 4 tensor (aas).

$$\begin{bmatrix} 0 & G_{yzzx} & 0 \\ G_{yzzx} & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

$$G_{yzzx} = 2\mathbb{G}_{xy}^{(4)}$$

* Rank 4 tensor (aaa).

$$\begin{bmatrix} 0 & G_{yzzx} & 0 \\ -G_{yzzx} & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

$$G_{yzzx} = -\mathbb{Q}_{z}^{(2)}$$

* Rank 4 tensor (sa).

$$\begin{bmatrix} 0 & 0 & G_{xxxy} \\ 0 & 0 & G_{yyxy} \\ 0 & 0 & G_{zzxy} \\ 0 & G_{yzzx} & 0 \\ G_{zxyz} & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

$$G_{xxxy} = -2\mathbb{G}_{xy}^{(5)} + \mathbb{Q}_{z}^{(3)} - \mathbb{Q}_{z}^{\alpha(2)} + \mathbb{Q}_{z}^{\beta(2)}$$
$$G_{yyxy} = 2\mathbb{G}_{xy}^{(5)} + \mathbb{Q}_{z}^{(3)} - \mathbb{Q}_{z}^{\alpha(2)} - \mathbb{Q}_{z}^{\beta(2)}$$
$$G_{zzxy} = \mathbb{Q}_{z}^{(3)} + 2\mathbb{Q}_{z}^{(4)} + 2\mathbb{Q}_{z}^{\alpha(2)}$$
$$G_{yzzx} = -\mathbb{G}_{xy}^{(5)} + \mathbb{Q}_{z}^{(4)} - \mathbb{Q}_{z}^{\alpha(2)} - \mathbb{Q}_{z}^{\beta(2)}$$
$$G_{zxyz} = \mathbb{G}_{xy}^{(5)} + \mathbb{Q}_{z}^{(4)} - \mathbb{Q}_{z}^{\alpha(2)} + \mathbb{Q}_{z}^{\beta(2)}$$

* Rank 4 tensor (as).

$$\begin{bmatrix} 0 & 0 & 0 & 0 & G_{yzzx} & 0 \\ 0 & 0 & 0 & G_{zxyz} & 0 & 0 \\ G_{xyxx} & G_{xyyy} & G_{xyzz} & 0 & 0 & 0 \end{bmatrix}$$

$$G_{yzzx} = \mathbb{G}_{xy}^{(6)} + \mathbb{Q}_{z}^{(6)} - \mathbb{Q}_{z}^{\alpha(3)} + \mathbb{Q}_{z}^{\beta(3)}$$
$$G_{zxyz} = -\mathbb{G}_{xy}^{(6)} + \mathbb{Q}_{z}^{(6)} - \mathbb{Q}_{z}^{\alpha(3)} - \mathbb{Q}_{z}^{\beta(3)}$$
$$G_{xyxx} = -2\mathbb{G}_{xy}^{(6)} + \mathbb{Q}_{z}^{(5)} - \mathbb{Q}_{z}^{\alpha(3)} + \mathbb{Q}_{z}^{\beta(3)}$$
$$G_{xyyy} = 2\mathbb{G}_{xy}^{(6)} + \mathbb{Q}_{z}^{(5)} - \mathbb{Q}_{z}^{\alpha(3)} - \mathbb{Q}_{z}^{\beta(3)}$$
$$G_{xyzz} = \mathbb{Q}_{z}^{(5)} + 2\mathbb{Q}_{z}^{(6)} + 2\mathbb{Q}_{z}^{\alpha(3)}$$

* Rank 4 tensor (s).

$$\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & G_{xxxy} & 0 & 0 & G_{xxyx} \\ 0 & 0 & 0 & 0 & 0 & G_{yyxy} & 0 & 0 & G_{yyyx} \\ 0 & 0 & 0 & 0 & 0 & G_{zzxy} & 0 & 0 & G_{zzyx} \\ 0 & 0 & 0 & 0 & G_{yzzx} & 0 & 0 & G_{yzxz} & 0 \\ 0 & 0 & 0 & G_{zxyz} & 0 & 0 & G_{zxzy} & 0 & 0 \\ G_{xyxx} & G_{xyyy} & G_{xyzz} & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

$$G_{xxxy} = \mathbb{G}_{4z}^{\alpha(1)} - \mathbb{G}_{4z}^{\beta(1)} + \mathbb{G}_{xy}^{(1)} + 2\mathbb{G}_{xy}^{(2)} + \mathbb{G}_{xy}^{(3)} - 2\mathbb{G}_{xy}^{(5)} + 2\mathbb{Q}_{z}^{(1)} + \mathbb{Q}_{z}^{(3)} - 2\mathbb{Q}_{z}^{\alpha(1)} - \mathbb{Q}_{z}^{\alpha(2)} + 2\mathbb{Q}_{z}^{\beta(1)} + \mathbb{Q}_{z}^{\beta(2)}$$
$$G_{xxyx} = \mathbb{G}_{4z}^{\alpha(1)} - \mathbb{G}_{4z}^{\beta(1)} + \mathbb{G}_{xy}^{(1)} + 2\mathbb{G}_{xy}^{(2)} + \mathbb{G}_{xy}^{(3)} + 2\mathbb{G}_{xy}^{(5)} + 2\mathbb{Q}_{z}^{(1)} - \mathbb{Q}_{z}^{(3)} - 2\mathbb{Q}_{z}^{\alpha(1)} + \mathbb{Q}_{z}^{\alpha(2)} + 2\mathbb{Q}_{z}^{\beta(1)} - \mathbb{Q}_{z}^{\beta(2)}$$
$$G_{yyxy} = -\mathbb{G}_{4z}^{\alpha(1)} - \mathbb{G}_{4z}^{\beta(1)} + \mathbb{G}_{xy}^{(1)} + 2\mathbb{G}_{xy}^{(2)} + \mathbb{G}_{xy}^{(3)} + 2\mathbb{G}_{xy}^{(5)} - 2\mathbb{Q}_{z}^{(1)} + \mathbb{Q}_{z}^{(3)} + 2\mathbb{Q}_{z}^{\alpha(1)} - \mathbb{Q}_{z}^{\alpha(2)} + 2\mathbb{Q}_{z}^{\beta(1)} - \mathbb{Q}_{z}^{\beta(2)}$$
$$G_{yyyx} = -\mathbb{G}_{4z}^{\alpha(1)} - \mathbb{G}_{4z}^{\beta(1)} + \mathbb{G}_{xy}^{(1)} + 2\mathbb{G}_{xy}^{(2)} + \mathbb{G}_{xy}^{(3)} - 2\mathbb{G}_{xy}^{(5)} - 2\mathbb{Q}_{z}^{(1)} - \mathbb{Q}_{z}^{(3)} + 2\mathbb{Q}_{z}^{\alpha(1)} + \mathbb{Q}_{z}^{\alpha(2)} + 2\mathbb{Q}_{z}^{\beta(1)} + \mathbb{Q}_{z}^{\beta(2)}$$
$$G_{zzxy} = 2\mathbb{G}_{4z}^{\beta(1)} + \mathbb{G}_{xy}^{(1)} + \mathbb{G}_{xy}^{(3)} + \mathbb{Q}_{z}^{(3)} + 2\mathbb{Q}_{z}^{(4)} + 2\mathbb{Q}_{z}^{\alpha(2)} - 4\mathbb{Q}_{z}^{\beta(1)}$$
$$G_{zzyx} = 2\mathbb{G}_{4z}^{\beta(1)} + \mathbb{G}_{xy}^{(1)} + \mathbb{G}_{xy}^{(3)} - \mathbb{Q}_{z}^{(3)} - 2\mathbb{Q}_{z}^{(4)} - 2\mathbb{Q}_{z}^{\alpha(2)} - 4\mathbb{Q}_{z}^{\beta(1)}$$
$$G_{yzzx} = 2\mathbb{G}_{4z}^{\beta(1)} + \mathbb{G}_{xy}^{(2)} - \mathbb{G}_{xy}^{(5)} - \mathbb{Q}_{z}^{(1)} + \mathbb{Q}_{z}^{(4)} - 4\mathbb{Q}_{z}^{\alpha(1)} - \mathbb{Q}_{z}^{\alpha(2)} - \mathbb{Q}_{z}^{\beta(2)}$$
$$G_{yzxz} = 2\mathbb{G}_{4z}^{\beta(1)} + \mathbb{G}_{xy}^{(2)} + \mathbb{G}_{xy}^{(5)} - \mathbb{Q}_{z}^{(1)} - \mathbb{Q}_{z}^{(4)} - 4\mathbb{Q}_{z}^{\alpha(1)} + \mathbb{Q}_{z}^{\alpha(2)} + \mathbb{Q}_{z}^{\beta(2)}$$
$$G_{zxyz} = 2\mathbb{G}_{4z}^{\beta(1)} + \mathbb{G}_{xy}^{(2)} + \mathbb{G}_{xy}^{(5)} + \mathbb{Q}_{z}^{(1)} + \mathbb{Q}_{z}^{(4)} + 4\mathbb{Q}_{z}^{\alpha(1)} - \mathbb{Q}_{z}^{\alpha(2)} + \mathbb{Q}_{z}^{\beta(2)}$$
$$G_{zxzy} = 2\mathbb{G}_{4z}^{\beta(1)} + \mathbb{G}_{xy}^{(2)} - \mathbb{G}_{xy}^{(5)} + \mathbb{Q}_{z}^{(1)} - \mathbb{Q}_{z}^{(4)} + 4\mathbb{Q}_{z}^{\alpha(1)} + \mathbb{Q}_{z}^{\alpha(2)} - \mathbb{Q}_{z}^{\beta(2)}$$
$$G_{xyxx} = \mathbb{G}_{4z}^{\alpha(1)} - \mathbb{G}_{4z}^{\beta(1)} + \mathbb{G}_{xy}^{(1)} + 2\mathbb{G}_{xy}^{(2)} - \mathbb{G}_{xy}^{(3)} - 2\mathbb{Q}_{z}^{(1)} + 2\mathbb{Q}_{z}^{\alpha(1)} - 2\mathbb{Q}_{z}^{\beta(1)}$$
$$G_{xyyy} = -\mathbb{G}_{4z}^{\alpha(1)} - \mathbb{G}_{4z}^{\beta(1)} + \mathbb{G}_{xy}^{(1)} + 2\mathbb{G}_{xy}^{(2)} - \mathbb{G}_{xy}^{(3)} + 2\mathbb{Q}_{z}^{(1)} - 2\mathbb{Q}_{z}^{\alpha(1)} - 2\mathbb{Q}_{z}^{\beta(1)}$$
$$G_{xyzz} = 2\mathbb{G}_{4z}^{\beta(1)} + \mathbb{G}_{xy}^{(1)} - \mathbb{G}_{xy}^{(3)} + 4\mathbb{Q}_{z}^{\beta(1)}$$

* Rank 4 tensor (a).

$$\begin{bmatrix} 0 & 0 & 0 & 0 & G_{yzzx} & 0 & 0 & G_{yzxz} & 0 \\ 0 & 0 & 0 & G_{zxyz} & 0 & 0 & G_{zxzy} & 0 & 0 \\ G_{xyxx} & G_{xyyy} & G_{xyzz} & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

$$G_{yzzx} = 2\mathbb{G}_{xy}^{(4)} + \mathbb{G}_{xy}^{(6)} - \mathbb{Q}_{z}^{(2)} + \mathbb{Q}_{z}^{(6)} - \mathbb{Q}_{z}^{\alpha(3)} + \mathbb{Q}_{z}^{\beta(3)}$$
$$G_{yzxz} = -2\mathbb{G}_{xy}^{(4)} + \mathbb{G}_{xy}^{(6)} + \mathbb{Q}_{z}^{(2)} + \mathbb{Q}_{z}^{(6)} - \mathbb{Q}_{z}^{\alpha(3)} + \mathbb{Q}_{z}^{\beta(3)}$$
$$G_{zxyz} = 2\mathbb{G}_{xy}^{(4)} - \mathbb{G}_{xy}^{(6)} + \mathbb{Q}_{z}^{(2)} + \mathbb{Q}_{z}^{(6)} - \mathbb{Q}_{z}^{\alpha(3)} - \mathbb{Q}_{z}^{\beta(3)}$$
$$G_{zxzy} = -2\mathbb{G}_{xy}^{(4)} - \mathbb{G}_{xy}^{(6)} - \mathbb{Q}_{z}^{(2)} + \mathbb{Q}_{z}^{(6)} - \mathbb{Q}_{z}^{\alpha(3)} - \mathbb{Q}_{z}^{\beta(3)}$$
$$G_{xyxx} = -2\mathbb{G}_{xy}^{(6)} + \mathbb{Q}_{z}^{(5)} - \mathbb{Q}_{z}^{\alpha(3)} + \mathbb{Q}_{z}^{\beta(3)}$$
$$G_{xyyy} = 2\mathbb{G}_{xy}^{(6)} + \mathbb{Q}_{z}^{(5)} - \mathbb{Q}_{z}^{\alpha(3)} - \mathbb{Q}_{z}^{\beta(3)}$$
$$G_{xyzz} = \mathbb{Q}_{z}^{(5)} + 2\mathbb{Q}_{z}^{(6)} + 2\mathbb{Q}_{z}^{\alpha(3)}$$

* Rank 4 tensor (t).

$$\begin{bmatrix} 0 & G_{xxxy} & 0 \\ G_{yyyx} & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & G_{zzxy} & 0 \\ G_{xxyx} & 0 & 0 \\ G_{yzzx} & 0 & 0 \\ 0 & 0 & 0 \\ 0 & G_{xyyy} & 0 \\ 0 & 0 & \frac{G_{xxxy}}{2} - \frac{G_{xxyx}}{2} + G_{xyyy} - G_{yyyx} + G_{zzxy} \end{bmatrix}$$

$$G_{xxxy} = \mathbb{G}_{4z}^{\alpha(1)} - \mathbb{G}_{4z}^{\beta(1)} + \mathbb{G}_{xy}^{(1)} + 2\mathbb{G}_{xy}^{(2)} + \mathbb{G}_{xy}^{(3)} - 2\mathbb{G}_{xy}^{(5)}$$
$$G_{yyyx} = -\mathbb{G}_{4z}^{\alpha(1)} - \mathbb{G}_{4z}^{\beta(1)} + \mathbb{G}_{xy}^{(1)} + 2\mathbb{G}_{xy}^{(2)} + \mathbb{G}_{xy}^{(3)} - 2\mathbb{G}_{xy}^{(5)}$$
$$G_{zzxy} = 2\mathbb{G}_{4z}^{\beta(1)} + \mathbb{G}_{xy}^{(1)} + \mathbb{G}_{xy}^{(3)}$$
$$G_{xxyx} = \mathbb{G}_{4z}^{\alpha(1)} - \mathbb{G}_{4z}^{\beta(1)} + \mathbb{G}_{xy}^{(1)} + 2\mathbb{G}_{xy}^{(2)} + \mathbb{G}_{xy}^{(3)} + 2\mathbb{G}_{xy}^{(5)}$$
$$G_{yzzx} = 2\mathbb{G}_{4z}^{\beta(1)} + \mathbb{G}_{xy}^{(2)} - \mathbb{G}_{xy}^{(5)}$$
$$G_{xyyy} = -\mathbb{G}_{4z}^{\alpha(1)} - \mathbb{G}_{4z}^{\beta(1)} + \mathbb{G}_{xy}^{(1)} + 2\mathbb{G}_{xy}^{(2)} - \mathbb{G}_{xy}^{(3)}$$